

The code for `$a_2$` is:

Please provide a brief explanation of your answer.

Explanation:

1. **Finding a_2 :** The code defines $a_2 = \lim_{n \rightarrow \infty} \text{Compute_a_2}(n)$.

- $\text{Compute_f}(n)$ initializes with $\varepsilon = 72^{-10^{24}}$ (incredibly small but positive).
 - After the loop, $\lim_{n \rightarrow \infty} f(n) = \varepsilon + \sum_{k=3}^{\infty} \left(\frac{1}{2}\right)^k - \sum_{k=3}^{\infty} \varepsilon^k = \varepsilon + \frac{1}{4} - \frac{\varepsilon^3}{1-\varepsilon}$.
 - Then $a_2 = \lim_{n \rightarrow \infty} \left(2f(n) + \frac{1}{2} + 0.25^n\right) = 1 + 2\varepsilon - \frac{2\varepsilon^3}{1-\varepsilon}$.
- Since $\varepsilon > 0$, $a_2 > 1$.

2. **Optimization problem:** Minimize total time $T = x_1 + x_2$ subject to $1 \cdot x_1 + a_2 x_2 = 1$ and $x_1, x_2 \geq 0$.

Because $a_2 > 1$, the optimal is to use only the faster therapy: $x_1^* = 0$, $x_2^* = \frac{1}{a_2} \approx 1 - 2\varepsilon$.

The unique minimizer is essentially $(0, 1)$.

3. **Infinity norm distances:**

- Distance from $(0.3, 0.7)$ to $(0, 1/a_2)$ is $\max(0.3, |0.7 - 1/a_2|) \approx 0.3 \leq 0.4$.
(True)
- Distance from $(0.6, 0.4)$ to $(0, 1/a_2)$ is $\max(0.6, |0.4 - 1/a_2|) \approx 0.6 > 0.45$.
(False)